Time Series Analysis Project

Moahhid Amir

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Section 1: Introduction

In this Time Series Analysis project I *projecting the* ***net ratings for the Los Angeles Lakers*** *for the remainder of the 2025-2026 NBA season* with a 95% prediction interval. I will do this using the net rating per game from the start of the season until now. This would allow me to predict the win percentage and standings for the Lakers for the rest of the season which in turn could give me insights as to who the lakers are most likely to face in the first round of the NBA playoffs.

Data Source: I used Basketball reference which has data available for every NBA game and player dating back to the 1950s.

Path: Basketball reference -> Team -> Game log -> Export data as Excel file.

Why Los Angeles Lakers?

The *Los Angeles Lakers* have had a great season so far and would like to close off the season strong.

Since two of the 3 superstars on the Los Angeles Lakers are out indefinitely for the remainder of the season, having a good understanding of the Lakers' net rating for the rest of the season would allow us to predict their win percentage and standings for the rest of the season which in turn could give us insights as to who they are most likely to face in the first round of the NBA playoffs.

About Los Angeles Lakers and Lebron James:

The Los Angeles Lakers are a professional basketball team based in Los Angeles, California. They are a member of the Western Conference Pacific Division of the National Basketball Association (NBA). The Lakers were founded in 1947 and have won 17 NBA championships, making them one of the most successful teams in NBA history. The team is known for its iconic purple and gold colors and has had many legendary players throughout its history, including Magic Johnson, Kareem Abdul-Jabbar, and Kobe Bryant.

Lebron James is a professional basketball player who currently plays for the Los Angeles Lakers and has played for the Cleveland Cavaliers and Miami Heat. He is widely regarded as one of the greatest basketball players of all time and has won four NBA championships, four NBA Most Valuable Player (MVP) awards, and two Olympic gold medals. Lebron James is known for his versatility, athleticism, and basketball IQ, and has been a dominant force in the NBA for over a 2 decade. He is on a run to win his 5th NBA championship this season, which would be a great way to end his career.

I collected game log data for every NBA team for the season and computed important metrics which allowed me to determine the Net/Offensive/Defensive ratings along with per game possessions for each game.

Using the following equations:

$$\text{Possessions} = 0.5\Big[(FGA - ORB + TOV + 0.44 * FTA) + (OppFGA - OppORB + OppTOV + 0.44 * OppFTA)\Big]$$

Note: The possession formula used is taken to be the average possession for the two teams in the game, this is because possession could be thought of as offensive and defensive possessions.

$$\text{ORtg} = 100 \times \frac{\text{Points Scored}}{\text{Possessions}}$$

$$\text{DRtg} = 100 \times \frac{\text{Points Allowed}}{\text{Possessions}}$$

$$\text{NetRtg} = \text{ORtg} - \text{DRtg}$$

The metrics are referenced from the book "Basketball on Paper" by Dean Oliver, published in 2004.

Since net rating is mostly roster and opponent dependent, we will be using rolling average of net rating for this time series analysis to smooth out the noise and capture the underlying structure of the time series. We will be using 1 week rolling/moving averages (~3 games) since this is a good balance between smoothing out the noise and not losing too much information about the underlying structure of the time series.

Now that we have created a log for every NBA team. Instead of manually assigning each data frame from the logged array, I will run attach() function so that I could easily access the data for each team.

```
# A tibble: 6 x 55
     Rk   Gtm Date                Column1 Opp   Rslt     Tm  Opp2 OT       FG
  <dbl> <dbl> <dttm>              <chr>   <chr> <chr> <dbl> <dbl> <chr> <dbl>
1     1     1 2025-10-21 00:00:00 <NA>    GSW   L       109   119 <NA>     42
2     2     2 2025-10-24 00:00:00 <NA>    MIN   W       128   110 <NA>     45
3     3     3 2025-10-26 00:00:00 @       SAC   W       127   120 <NA>     39
4     4     4 2025-10-27 00:00:00 <NA>    POR   L       108   122 <NA>     40
5     5     5 2025-10-29 00:00:00 @       MIN   W       116   115 <NA>     45
6     6     6 2025-10-31 00:00:00 @       MEM   W       117   112 <NA>     39
# i 45 more variables: FGA <dbl>, `FG%` <dbl>, `3P` <dbl>, `3PA` <dbl>,
#   `3P%` <dbl>, `2P` <dbl>, `2PA` <dbl>, `2P%` <dbl>, `eFG%` <dbl>, FT <dbl>,
#   FTA <dbl>, `FT%` <dbl>, ORB <dbl>, DRB <dbl>, TRB <dbl>, AST <dbl>,
#   STL <dbl>, BLK <dbl>, TOV <dbl>, PF <dbl>, FG3 <dbl>, FGA4 <dbl>,
#   `FG%5` <dbl>, `3P6` <dbl>, `3PA7` <dbl>, `3P%8` <dbl>, `2P9` <dbl>,
#   `2PA10` <dbl>, `2P%11` <dbl>, `eFG%12` <dbl>, FT13 <dbl>, FTA14 <dbl>,
#   `FT%15` <dbl>, ORB16 <dbl>, DRB17 <dbl>, TRB18 <dbl>, AST19 <dbl>, ...
```

```
tibble [77 x 55] (S3: tbl_df/tbl/data.frame)
 $ Rk         : num [1:77] 1 2 3 4 5 6 7 8 9 10 ...
 $ Gtm        : num [1:77] 1 2 3 4 5 6 7 8 9 10 ...
 $ Date       : POSIXct[1:77], format: "2025-10-21" "2025-10-24" ...
 $ Column1    : chr [1:77] NA NA "@" NA ...
```

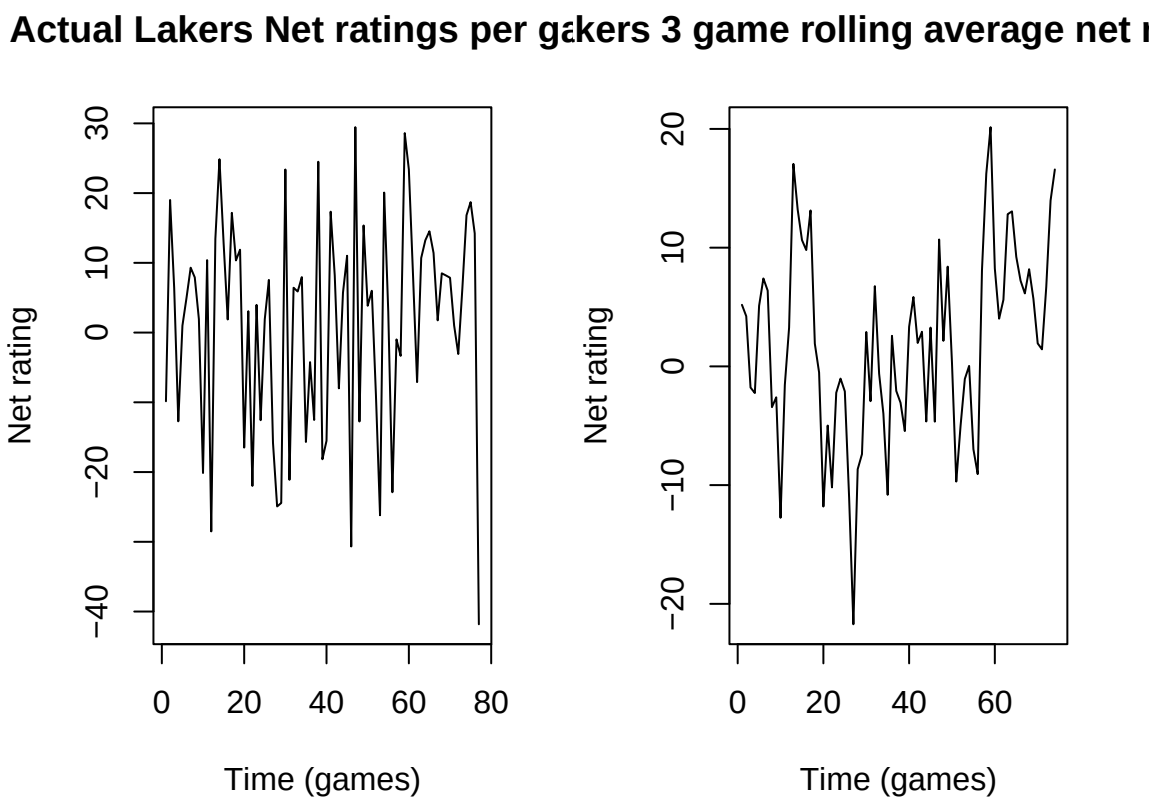
```
$ Opp        : chr [1:77] "GSW" "MIN" "SAC" "POR" ...
$ Rslt       : chr [1:77] "L" "W" "W" "L" ...
$ Tm         : num [1:77] 109 128 127 108 116 117 130 123 118 102 ...
$ Opp2       : num [1:77] 119 110 120 122 115 112 120 115 116 122 ...
$ OT         : chr [1:77] NA NA NA NA ...
$ FG         : num [1:77] 42 45 39 40 45 39 48 50 37 37 ...
$ FGA        : num [1:77] 77 76 80 86 83 76 95 85 83 81 ...
$ FG%        : num [1:77] 0.545 0.592 0.488 0.465 0.542 0.513 0.505 0.588 0.446 0.457 .
$ 3P         : num [1:77] 8 12 8 7 11 13 15 9 10 12 ...
$ 3PA        : num [1:77] 32 29 25 27 31 38 46 23 31 30 ...
$ 3P%        : num [1:77] 0.25 0.414 0.32 0.259 0.355 0.342 0.326 0.391 0.323 0.4 ...
$ 2P         : num [1:77] 34 33 31 33 34 26 33 41 27 25 ...
$ 2PA        : num [1:77] 45 47 55 59 52 38 49 62 52 51 ...
$ 2P%        : num [1:77] 0.756 0.702 0.564 0.559 0.654 0.684 0.673 0.661 0.519 0.49 ..
$ eFG%       : num [1:77] 0.597 0.671 0.538 0.506 0.608 0.599 0.584 0.641 0.506 0.531 .
$ FT         : num [1:77] 17 26 41 21 15 26 19 14 34 16 ...
$ FTA        : num [1:77] 28 32 46 26 22 33 23 16 44 20 ...
$ FT%        : num [1:77] 0.607 0.813 0.891 0.808 0.682 0.788 0.826 0.875 0.773 0.8 ...
$ ORB        : num [1:77] 7 7 10 12 11 5 8 8 12 13 ...
$ DRB        : num [1:77] 32 31 37 30 29 35 33 26 26 34 ...
$ TRB        : num [1:77] 39 38 47 42 40 40 41 34 38 47 ...
$ AST        : num [1:77] 23 26 26 23 34 24 33 29 23 23 ...
$ STL        : num [1:77] 7 6 9 11 6 8 14 11 8 8 ...
$ BLK        : num [1:77] 2 3 3 5 5 3 2 5 5 5 ...
$ TOV        : num [1:77] 20 12 17 25 19 15 12 17 14 20 ...
$ PF         : num [1:77] 21 20 17 19 22 16 26 22 30 11 ...
$ FG3        : num [1:77] 38 41 45 47 41 41 43 42 36 48 ...
$ FGA4       : num [1:77] 78 88 101 103 82 93 84 85 74 93 ...
$ FG%5       : num [1:77] 0.487 0.466 0.446 0.456 0.5 0.441 0.512 0.494 0.486 0.516 ...
$ 3P6        : num [1:77] 17 13 18 14 15 14 9 9 13 16 ...
$ 3PA7       : num [1:77] 40 38 43 43 30 45 33 40 33 39 ...
$ 3P%8       : num [1:77] 0.425 0.342 0.419 0.326 0.5 0.311 0.273 0.225 0.394 0.41 ...
$ 2P9        : num [1:77] 21 28 27 33 26 27 34 33 23 32 ...
$ 2PA10      : num [1:77] 38 50 58 60 52 48 51 45 41 54 ...
$ 2P%11      : num [1:77] 0.553 0.56 0.466 0.55 0.5 0.563 0.667 0.733 0.561 0.593 ...
$ eFG%12     : num [1:77] 0.596 0.54 0.535 0.524 0.591 0.516 0.565 0.547 0.574 0.602 ..
$ FT13       : num [1:77] 26 15 12 14 18 16 25 22 31 10 ...
$ FTA14      : num [1:77] 29 21 18 15 22 19 36 30 40 11 ...
$ FT%15      : num [1:77] 0.897 0.714 0.667 0.933 0.818 0.842 0.694 0.733 0.775 0.909 .
$ ORB16      : num [1:77] 9 15 13 19 9 13 9 15 8 7 ...
$ DRB17      : num [1:77] 31 25 27 30 25 29 33 23 29 30 ...
$ TRB18      : num [1:77] 40 40 40 49 34 42 42 38 37 37 ...
$ AST19      : num [1:77] 29 24 27 24 30 28 29 23 22 37 ...
$ STL20      : num [1:77] 10 4 13 12 8 9 7 13 7 13 ...
$ BLK21      : num [1:77] 4 3 4 9 5 0 3 3 1 6 ...
```

```
$ TOV22      : num [1:77] 19 12 16 19 18 11 15 18 19 11 ...
$ PF23       : num [1:77] 27 27 26 17 23 25 20 18 36 14 ...
$ Possessions: num [1:77] 101.5 94.7 109.6 110 100.7 ...
$ Ortg       : num [1:77] 107.3 135.2 115.9 98.2 115.2 ...
$ Drtg       : num [1:77] 117 116 110 111 114 ...
$ NRtg       : num [1:77] -9.848 19.015 6.388 -12.725 0.993 ...
```

Section 2 : Modeling

For our modeling, we will first take rolling average as stated in Section 1, then try to understand the underlying structure of our time series, applying time series modeling techniques to fit a good model for our data, and finally perform forecasting for the rest of the season.

Take the 1 week rolling averages.

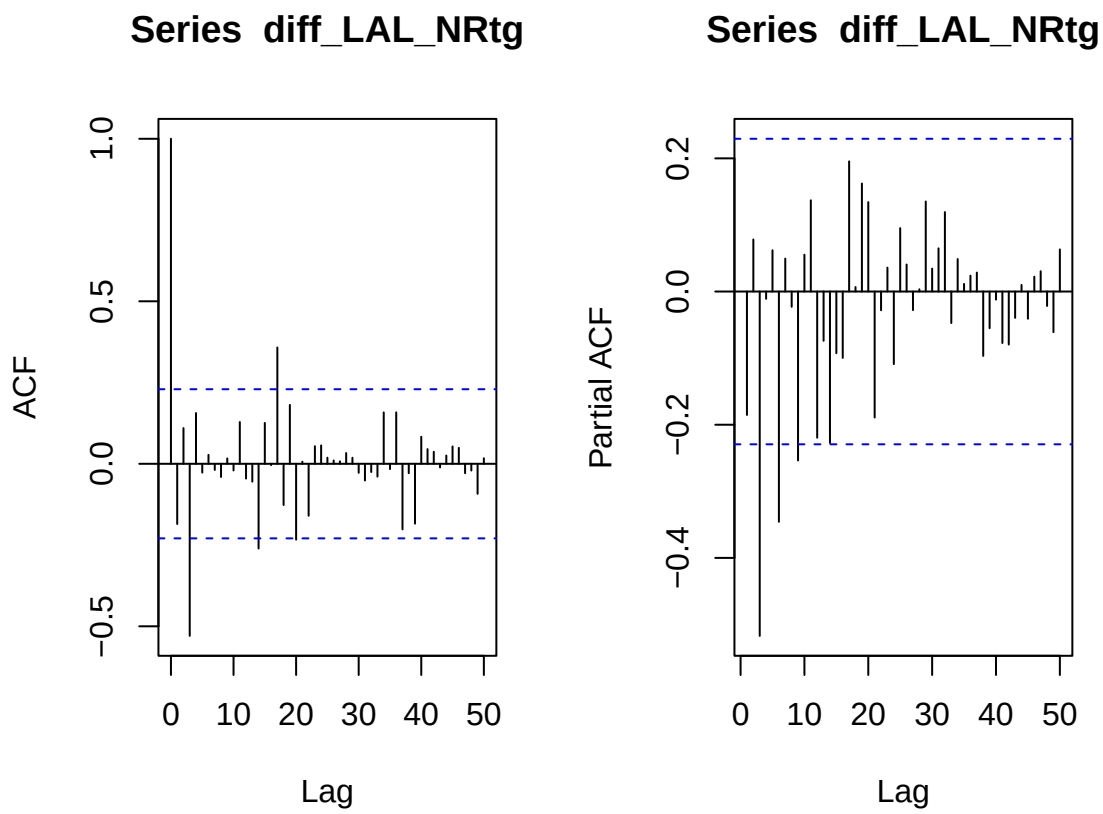


From the plot of our time series, we observe high volatility along with heteroscedasticity (not too obvious), which is expected given that we are dealing with net ratings which is highly volatile by nature.

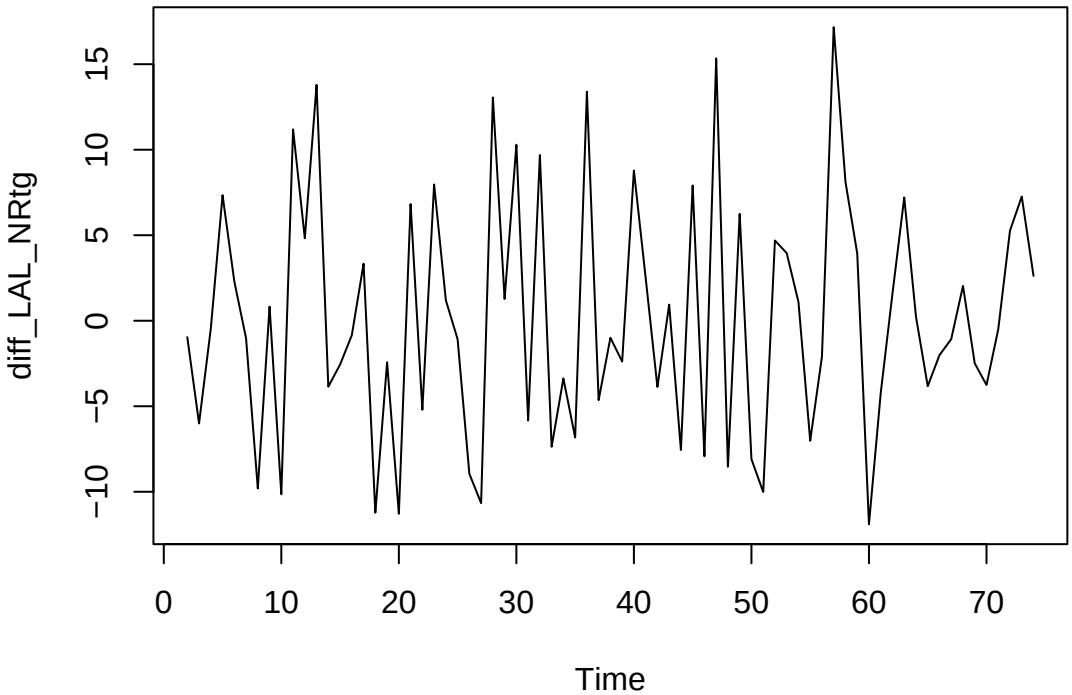
```
Augmented Dickey-Fuller Test
```

```
data:  LAL_rolling_NRtg
Dickey-Fuller = -2.7825, Lag order = 4, p-value = 0.2564
alternative hypothesis: stationary
```

The augmented dickey-fuller test gives us a p-value of 0.2129 which implies that we have insignificant evidence to reject the Null hypothesis that the series has a unit root (implying non-stationarity). So I will take the first difference of this series to try an achieve stationarity.




```
    Augmented Dickey-Fuller Test

data:  diff_LAL_NRtg
Dickey-Fuller = -4.3446, Lag order = 4, p-value = 0.01
alternative hypothesis: stationary
```



After we take the difference, we get the augmented dickey-fuller test p-value of 0.01 which implies that we have significant evidence to reject the Null hypothesis that the series has a unit root (implying stationarity).

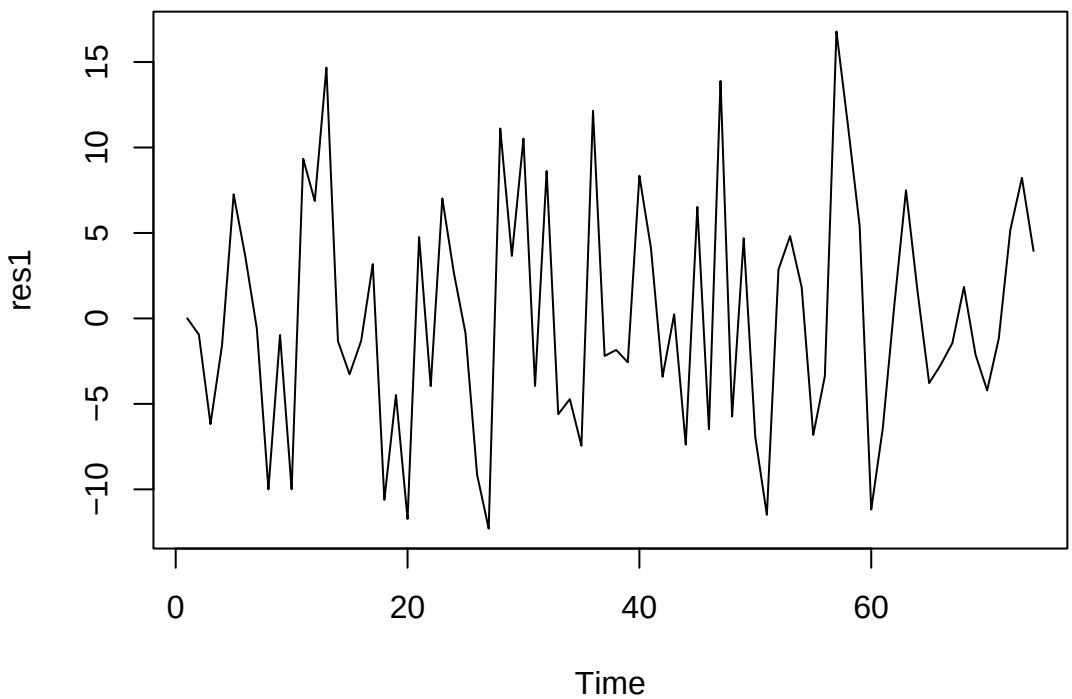
We can see that the ACF plot for non-seasonally differenced ($d = 1$) series shows spikes at lag 3, 17 and at 39 (not too obvious) but slowly decays to 0, whereas the PACF plot shows a significant spike at lag 1 and no structural significance beyond this. I will try to fit an ARIMA(1,1,0) model to this data to see if it is a good fit, then perform model diagnosis to check adequacy; possibly overfitting the model if necessary.

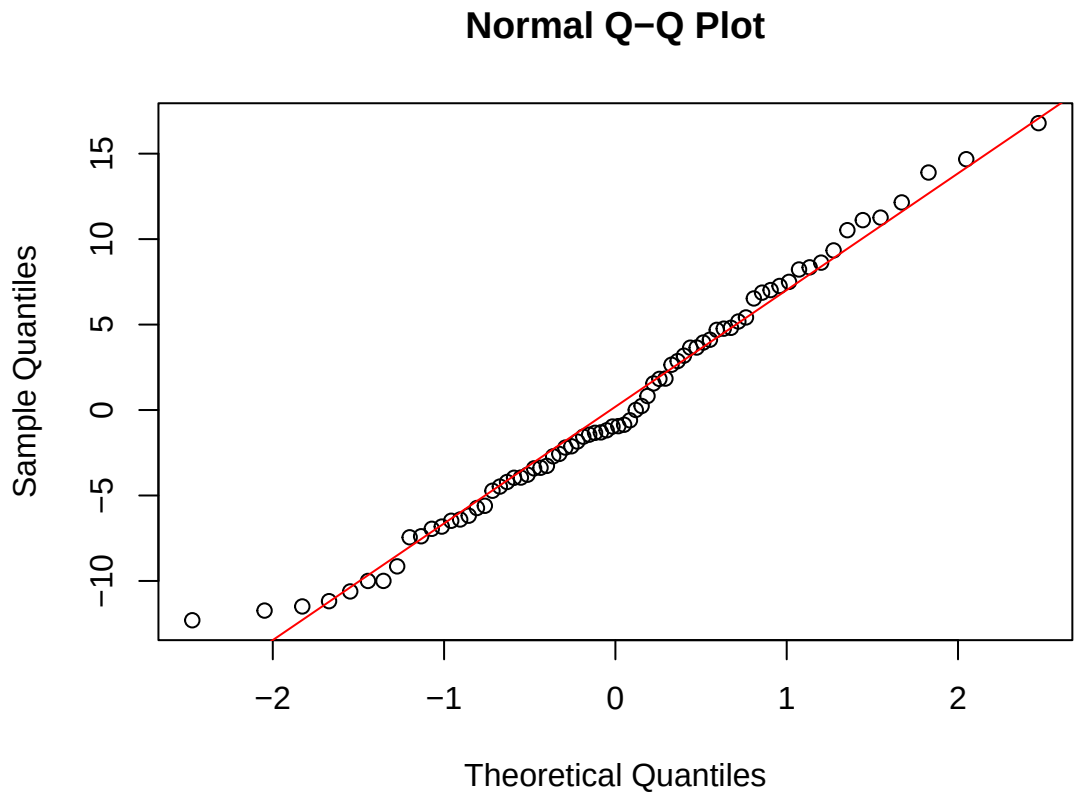
So we fit a ARIMA model with $(p, d, q) = (1, 1, 0)$ and $(P, D, Q) = (0, 0, 0)$ with no seasonal trend observed.

```
Series: LAL_rolling_NRtg
ARIMA(1,1,0)

Coefficients:
          ar1
      -0.1827
s.e.   0.1143

sigma^2 = 48.26:  log likelihood = -244.59
```

```
AIC=493.19   AICc=493.36   BIC=497.77

Training set error measures:
                    ME    RMSE      MAE      MPE     MAPE      MASE       ACF1
Training set 0.1760454 6.85255 5.642768 70.03012 232.6522 0.9876152 0.01124643
```

Now that we have fit the model, let us perform residual analysis for model diagnostics to check the adequacy of our model.






```
    Box-Ljung test

data:  res1
X-squared = 86.473, df = 52, p-value = 0.001893
```

From our model diagnostics, namely, residual analysis, we observe that the residuals comply with many assumptions required to proceed. The residuals do not show any significant autocorrelation structure (confirmed using Ljung Box test) with the plot of our residuals also exhibit white noise like behavior.

Try and overfit the model by increasing the AR order by 1 to see if we can get a better fit, but we want to avoid overfitting which would lead to poor forecasting performance.

```
Series: LAL_rolling_NRtg
ARIMA(2,1,0)

Coefficients:
          ar1     ar2
      -0.1676  0.0776
s.e.   0.1161  0.1168
```

```
sigma^2 = 48.64:  log likelihood = -244.37
AIC=494.75   AICc=495.09   BIC=501.62

Training set error measures:
                    ME     RMSE      MAE      MPE     MAPE     MASE       ACF1
Training set 0.1727667 6.831391 5.652743 56.15748 218.0976 0.989361 0.03707582
```

Notice that the second AR term is not significant and the AIC of this model is higher than the previous model, so we will stick with our first model which is ARIMA(1,1,0).

Since the model has captured the underlying structure of our time series to some extent, we could proceed with forecasting, but before doing so, let us perform cross validation to check the accuracy of our model.

We will use rolling forecast method to perform cross validation for performance check.

```
# A tsibble: 1,995 x 3 [1]
# Key:       .id [35]
   index  value   .id
   <dbl>  <dbl> <int>
 1     1   5.19     1
 2     2   4.23     1
 3     3  -1.78     1
 4     4  -2.24     1
 5     5   5.10     1
 6     6   7.41     1
 7     7   6.38     1
 8     8  -3.43     1
 9     9  -2.60     1
10    10 -12.8      1
# i 1,985 more rows
```

```
# A tibble: 1 x 10
  .model .type    ME  RMSE   MAE   MPE  MAPE  MASE RMSSE  ACF1
  <chr>  <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 arima  Test  0.530  6.76  5.48  108.  319. 0.959 0.962 0.106
```

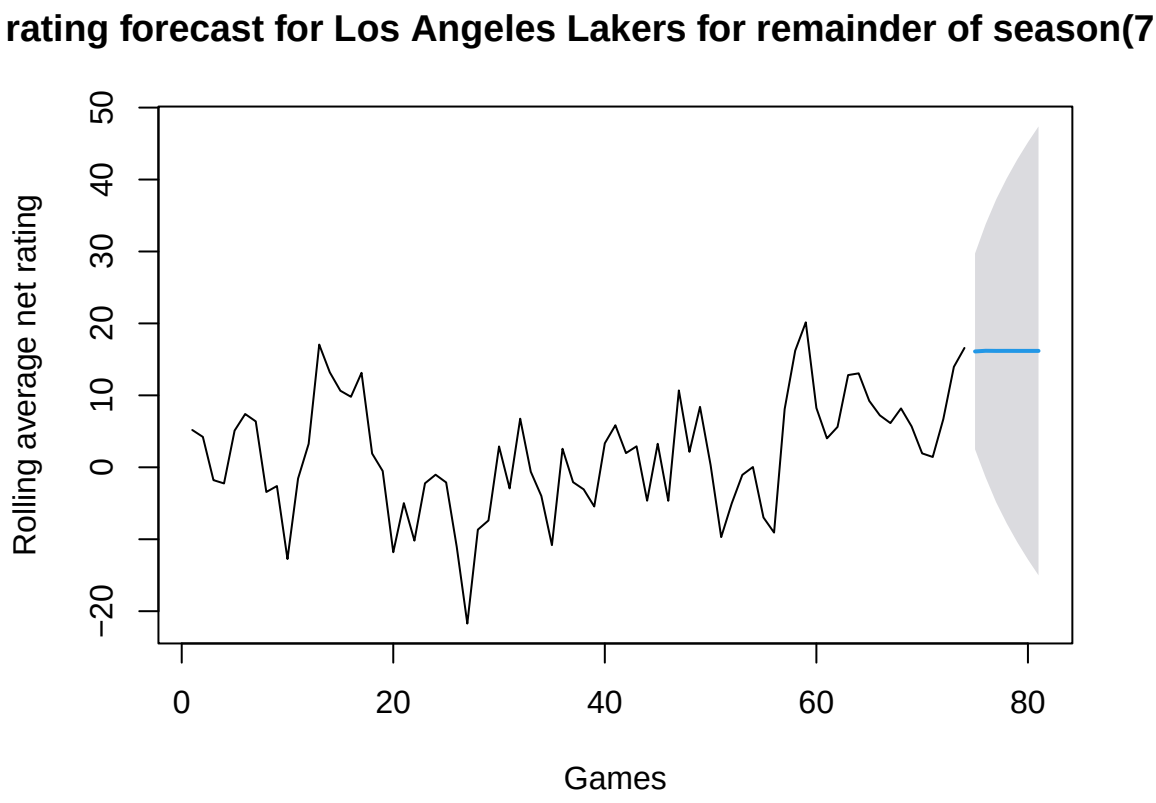
Above we have used rolling forecast cross validation to evaluate the performance of our model, initializing 40 games which is roughly half of the season (rolling average rating) to train our model with, then using 1 step ahead rolling forecast to evaluate the performance. From the accuracy metrics, we observe that the MASE is 0.995 which is relatively high, but good enough given the high volatility of our stock price data.

Section 3.1 : Results Part 1 (Forecasting)

From the modeling of our time series in section 2 we can see that the ARIMA(1,1,0) model is a good fit for our data. The residuals of the model show no significant autocorrelation

and the ACF and PACF plots of the residuals show no significant spikes, indicating that the model has captured the underlying structure of the data well.

So we can proceed to our forecasting step.



The forecast shows a relatively flat line which is the typical behavior observed in ARIMA models for highly volatile data especially if the best model to explain the underlying structure of our data is AR(1) with MLE being close to 0. The 95% prediction interval is quite wide which is expected given the high volatility of our Net rating.

Section 3.2 : Results Part 2 (Table)

The forecasted net ratings for the next 7 games are as follows:

Game	Forecasted NRtg	95% PI Lower	95% PI Upper	Opponent	Opponent Avg NRtg
76	0.871	-13.340	15.082	Cleveland Cavaliers (East)	4.3
77	0.119	-18.114	18.351	Oklahoma City Thunder (West)	11.6
78	0.266	-21.545	22.077	Dallas Mavericks (West)	-5.2

Game	Forecasted NRtg	95% PI Lower	95% PI Upper	Opponent	Opponent Avg NRtg
79	0.237	-24.591	25.066	Oklahoma City Thunder (West)	11.6
80	0.243	-27.283	27.769	Golden State Warriors (West)	-0.3
81	0.242	-29.738	30.222	Phoenix Suns (West)	1.5
82	0.242	-32.006	32.490	Utah Jazz (West)	-8.6

Above I have also listed out the opponent whom the Los Angeles Lakers are facing for the next 7 games along with their current average net rating for the season; given each team continues to perform as they are now.

Note: Opponent net ratings were retrieved from Advanced Statistics listed as the average net rating per 100 possessions.

Section 4 : Conclusion

In conclusion, we have successfully modeled the net rating of the Los Angeles Lakers using an ARIMA(1,1,0) model and performed forecasting for the remainder of the season. The forecasted net ratings for the next 7 games show a relatively flat line with wide prediction intervals, which is expected given the high volatility of our data.

Since the end goal was to understand where the lakers are most likely to finish in the standings, we made a fair assumption that all other Western Conference teams maintain their current average net rating for the remainder of the season.

From the prediction interval in the table we observe that the prediction intervals grow wider the further away we try to forecast the net rating, this tells us the true nature of uncertainty of NBA games outcomes, especially given the injury status of the Lakers' roster, and what other teams are dealing with. From the table, we also observe that non of the opponent average Net rating for the next 7 games is within the 95% prediction interval of the forecasted net rating for the Lakers, which does not allow us to make certain conclusions about the Lakers' win/loss outcomes for the remaining games, but we could use the forecasted Net rating for the lakers, and the average net rating for the opponents to make a fair assumption about the Lakers' win record for the next 7 games as follows:

Game	Outcome
Game 1	Lose
Game 2	Lose
Game 3	Win
Game 4	Lose
Game 5	Win
Game 6	Lose

Game	Outcome
Game 7	Win

Final record: 52-30

Assuming each team wins games proportional to their win percentage as of now, we could project the final standings for the west to be as follows:

Team	Record	WinPCT
OKC	65-17	0.793
SAS	62-20	0.756
DEN	53-29	0.646
LAL	52-30	0.634
HOU	51-31	0.622
MIN	49-33	0.598
PHO	45-37	0.549
LAC	43-39	0.524

From the above table, we can see that the Lakers are most likely to finish 4th in the Western Conference standings, which would mean they would face the 5th seed in the first round of the playoffs, which is most likely to be the Houston Rockets given their current record and net rating.

A significant limitation of using the ARIMA model here is that we assume future results using past data, this becomes a problem when we are dealing with sports data since there are so many other factors to be taken into consideration which may cause limitation in the accuracy of our forecasts.

Key limitations:

- Injury status
- Roster changes (starting lineup)
- Home/Away games
- Opponent

Extentions

To extend our SARIMA model, we could incorporate predictor variables such as home/away games, injury status of key players, etc. This would allow us to get a more nuanced understanding of the Lakers' chances of winning each game and could potentially lead to better predictions for the remainder of the season. And what this means for Lebron James and the Lakers given two of the three superstars are out indefinitely for the remainder of the season.

For an extention we could use logistic regression to model the probability of the contending teams to predict the win/loss record using average net rating for the opponents, along with other features such as home/away games, injury status of key players, etc. This would allow us to get a more nuanced understanding of the Lakers' chances of winning each game and could potentially lead to better predictions for the remainder of the season.

References

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